

ANN3: Perceptron (Even-Odd) – Detailed Code & Viva Explanation

Cell 1: Import Library

Code:

```
import numpy as np
```

Explanation (what to say in viva):

Here I am importing numpy for numerical operations.

It helps in performing vector and matrix calculations required in neural networks.

Cell 2: Define Dataset

Code:

```
x = np.array([[0],[1],[2],[3],[4],[5],[6],[7],[8],[9]])  
y = np.array([0,1,0,1,0,1,0,1,0,1])
```

Explanation (what to say in viva):

X contains input numbers from 0 to 9.

y contains labels where 0 means even and 1 means odd.

This is supervised learning because outputs are already known.

Cell 3: Initialize Parameters

Code:

```
w = np.zeros(1)  
b = 0  
alpha = 0.1  
epochs = 10
```

Explanation (what to say in viva):

w is initialized to zero as starting weight.

b is bias which shifts decision boundary.

alpha is learning rate controlling update step.

epochs represent number of training iterations.

Cell 4: Activation Function

Code:

```
def step(z):  
    if z >= 0:  
        return 1  
    else:  
        return 0
```

Explanation (what to say in viva):

This is step activation function.

If input is positive output is 1, otherwise 0.

It converts continuous value into binary output.

Cell 5: Training Loop

Code:

```
for _ in range(epochs):
    for i in range(len(X)):
        z = np.dot(X[i], w) + b
        y_hat = step(z)
        w = w + alpha * (y[i] - y_hat) * X[i]
        b = b + alpha * (y[i] - y_hat)
```

Explanation (what to say in viva):

This loop trains the perceptron.

z is calculated as weighted sum of input and bias.

y_hat is predicted output using step function.

Error is (y - y_hat).

We update weights and bias using perceptron learning rule.

This helps model learn correct classification.

Cell 6: Prediction

Code:

```
num = int(input())
z = np.dot([num % 2], w) + b
result = step(z)
```

Explanation (what to say in viva):

User enters a number.

num % 2 converts number into 0 (even) or 1 (odd).

Prediction is calculated using trained model.

Result gives classification.

Cell 7: Output

Code:

```
if result == 0:
    print('even')
else:
    print('odd')
```

Explanation (what to say in viva):

This prints final classification.

If output is 0 → even number.

If output is 1 → odd number.